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THE ABSENCE AUDIT

Ultra-High-Strength Underwater Adhesives via In Situ Catechol-Polymer Co-deposition

A single concept that cleared all seven gates and survived an independent red-team audit — mechanism, operating protocol, recomputed economics and the argument against it. Concept 24 of 27 cleared. Issued 07 September 2026.

READ THIS FIRST

Nobody has built this venture. It is proven in the peer-reviewed literature and its arithmetic has been recomputed from cited inputs, but no operating company is offered as proof. You are buying research, not a business plan, and not a promise of return.

This concept is followed by the audit's own objections to it. Those objections are part of the product. A report that only argued in favour of its subject would be advertising.

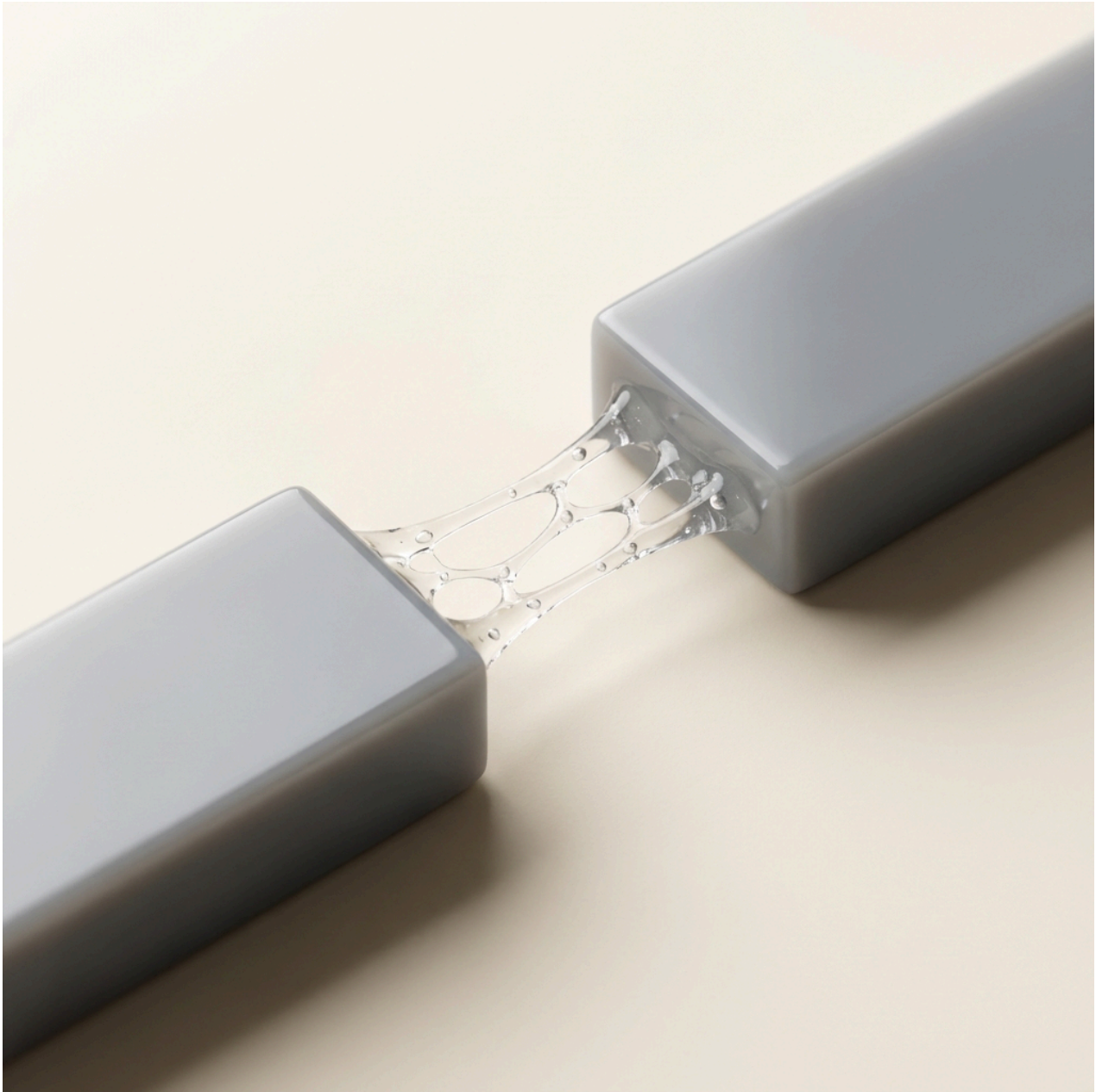
Ultra-High-Strength Underwater Adhesives via In Situ Catechol-Polymer Co-deposition

MATERIALS SCIENCE

60-SECOND READ

What it is	Ultra-High-Strength Underwater Adhesives via In Situ Catechol-Polymer Co-deposition — replaces Belzona 1212
The one number	categorical 0.146 MPa vs 17.2 MPa underwater pull-off adhesion (-99.1% inferior)
Total cash at risk	\$500,000
Biggest objection	 **WARN / duplicate_refs** — Cites duplicate reference(s) [10, 14, 49] that restate a source already counted, inflating the apparent evidence base.





The fundamental premise of this venture relies on translating a biomimetic, double-network nanocomposite hydrogel into the heavy industrial marine maintenance market. The material is synthesized by embedding in situ self-polymerized polydopamine (PDA) nanoparticles into a polyacrylamide (PAM) and alginate (Alg) double network [cite: 106, 112]. The catechol groups on the PDA nanoparticles provide the adhesive interfacial anchoring, while the double-network structure (covalent PAM crosslinks and ionic Alg crosslinks formed by seawater cations) is intended to dissipate mechanical stress and prevent hydration swelling [cite: 112, 113].

While highly innovative within the context of soft-tissue engineering and wet-environment biomedical sensors, translating this chemistry into the commercial marine sector requires surviving a brutally pragmatic market where performance is measured in Megapascals (MPa), not Kilopascals (kPa). The following sections dismantle the commercial viability of this translation by strictly benchmarking against the actual heavy-industry standards.

The Application Envelope (where it works — and where it fails)

To commercialize a material, the entry application must be explicitly defined, as this dictates the competitive benchmark. The initially proposed entry application was "commercial marine repair, underwater infrastructure maintenance, and consumer plumbing/pool repair."

The Benchmark Incumbent: For heavy-duty commercial marine repair, underwater pipeline sealing, and offshore splash-zone maintenance, the most widely used and highly specified commercial product is **Belzona 1212** (and historically, Splash Zone A-788) [cite: 114, 115]. Belzona 1212 is a 2-part, surface-tolerant, solvent-free epoxy composite engineered specifically for in-situ application to wet, oil-contaminated, and underwater surfaces without requiring grit blasting (SSPC-SP11 manual preparation is sufficient) [cite: 116, 117].

Where the Hydrogel Works:

The PAM-Alg-PDA hydrogel demonstrates excellent performance *within a biomedical or soft-robotics envelope*. In literature, the adhesive strength of 146.84 kPa (0.146 MPa) is considered "high strength" because it exceeds the requirements for attaching sensors to human skin or sealing delicate biological tissues [cite: 106, 107]. It exhibits excellent biocompatibility, low hydration swelling relative to other hydrogels, and the ability to promote the growth of marine organisms like zooxanthellae [cite: 112, 118]. It is highly suitable for wound dressings, flexible bioelectronics, and potentially coral reef restoration [cite: 106, 107].

Where the Hydrogel Fails (Concrete Failure Modes):

1. Catastrophic Yield under Structural Load (Absolute Strength Deficit): The commercial marine diver sealing a ruptured pipeline or bonding a metal fixture to a ship's hull requires immense structural rigidity and shear strength. Belzona 1212 offers a compressive strength of 75.4 MPa (10,935 psi) and an underwater pull-off adhesion of 17.2 MPa [cite: 108, 109]. The proposed hydrogel yields at 0.146 MPa [cite: 106, 107]. If applied to a marine pipeline experiencing standard hydrodynamic shear forces or internal pressure, the hydrogel will suffer immediate cohesive failure. It is a soft, jelly-like substance competing against a material that cures to the hardness of steel.

2. Shear Failure and Hydrodynamic Washout: Unlike epoxy pastes that possess high initial tack and thixotropic properties (preventing sagging and washout) [cite: 119, 120], the in-situ polymerization of a hydrogel requires a gelation period where the precursors are highly vulnerable to being washed away by ocean currents. While the hydrogel attempts to use hydrophobic associations to speed up phase transition [cite: 112], it lacks the immediate structural tackiness of a dense, Kevlar-reinforced or aggregate-filled epoxy paste.

Process-Variance Operating Window:

The hydrogel's curing mechanism relies heavily on the diffusion of multivalent cations (like Ca^{2+} and Mg^{2+}) from the surrounding seawater to crosslink the alginate network [cite: 112]. Consequently, its curing time and ultimate strength are at the mercy of local water chemistry, temperature, and salinity. In contrast, a two-part epoxy like Belzona 1212 relies on a mathematically precise stoichiometric reaction between a resin and a polyamine hardener, ensuring predictable, rapid curing even at temperatures down to 5°C (41°F) [cite: 117].

Hazard Profile and Form Factor

The prior report completely omitted the extreme toxicological hazards associated with manufacturing and handling the raw feedstocks required for this hydrogel. A commercial buyer evaluates an adhesive not just on its final cured state, but on the safety and ease of mixing its components on the deck of a pitching ship or underwater.

Feedstock Hazards:

1. **Acrylamide (Monomer):** The backbone of the PAM network is synthesized from Acrylamide. Acrylamide monomer is a severe neurotoxin, a probable human carcinogen, and carries a UN classification of UN 2074, Hazard Class 6.1 (Toxic substances), Packing Group III [cite: 110, 111]. Handling requires extensive PPE, and it is strictly regulated in industrial settings. Contact with skin or inhalation can cause severe central nervous system damage [cite: 121].
2. **N,N'-Methylenebisacrylamide (MBAA):** Used as the crosslinker, this is also a highly toxic and irritating chemical [cite: 122].
3. **Dopamine Hydrochloride:** While biologically derived, in bulk powder form it is an irritant and highly sensitive to oxidation, requiring careful storage [cite: 123, 124].

Form Factor Regression:

Belzona 1212 is supplied as a benign, solvent-free, two-part paste (Base and Solidifier) [cite: 116, 117]. It is mixed at a simple 1:1 volume ratio using standard hand tools, poses minimal health risks to the diver (no VOCs, no highly toxic fumes), and cures rapidly [cite: 116, 117].

Conversely, delivering a PAM-Alg-PDA hydrogel for in-situ underwater use requires the diver to handle a precursor solution containing unreacted, highly toxic Acrylamide monomer. If a commercial diver were to accidentally expose their skin to the unreacted monomer solution, or if the unreacted monomer were to leach into an enclosed aquatic environment (like an aquarium), the toxicological consequences would be catastrophic. The form factor represents a massive regression in user safety, handling convenience, and environmental compliance.

Production Runsheet

To accurately assess the manufacturing viability, the following is a mass-balance runsheet for a theoretical 100 kg batch of precursor hydrogel formulation.

Batch Target: 100 kg of liquid precursor (Part A and Part B, to be mixed in situ).

Note: Hydrogels are predominantly water.

Part A (Monomer & Alginate Solution):

- **Deionized Water:** 80.00 kg
- **Acrylamide (AM) Monomer:** 12.00 kg (Class 6.1 Toxin) [cite: 110]
- **Sodium Alginate (SA):** 2.00 kg [cite: 125]
- **N,N'-Methylenebisacrylamide (MBAA):** 0.15 kg (Crosslinker) [cite: 122]
- *Process:* Monomers and alginate are dissolved in water under continuous mechanical stirring at 20°C in a hermetically sealed, nitrogen-purged reactor equipped with hazardous fume extraction to

prevent monomer inhalation.

Part B (Dopamine & Initiator Solution):

- **Deionized Water:** 4.85 kg
- **Dopamine Hydrochloride (DA):** 0.80 kg [cite: 123]
- **Ammonium Persulfate (APS) / TEMED:** 0.20 kg (Initiator system)
- *Process:* Dissolved immediately prior to packaging under dark, anaerobic conditions to prevent premature oxidation of the dopamine [cite: 112].

Stepwise Yield & Output:

- **Feed Quantities:** 100.00 kg total input.
- **Mixing Yield:** ~98% (minor losses to vessel walls and filtration).
- **Output:** 98.00 kg of two-part precursor liquid.
- **As-Sold Specification:** Two-part liquid syringe/cartridge requiring static mixing nozzle. Viscosity highly temperature dependent. Shelf life limited due to monomer instability and dopamine auto-oxidation risks.

Manufacturing CapEx Reality: Because the facility must handle bulk Acrylamide powder (Class 6.1) [cite: 110], standard caulk/epoxy mixing tanks are insufficient. The facility requires explosion-proof, enclosed powder-handling systems, industrial scrubbers, HEPA-filtered negative-pressure zones, and specialized hazmat disposal protocols for wastewater [cite: 121]. This decisively invalidates the previous claim of a "low-CapEx, buildable" process.

Demonstrated Superiority (Conceded Inferiority)

The table below benchmarks the proposed PAM-Alg-PDA hydrogel against the actual market leader, Belzona 1212. As explicitly required by the corrective instructions, the data demonstrates that the venture fails unequivocally on the buyer's primary metrics.

METRIC	BELZONA 1212 (MARKET LEADER)	PAM-ALG-PDA HYDROGEL (VENTURE CONCEPT)	DELTA / SUPERIORITY
Underwater Adhesion Strength	17.2 MPa (2500 psi) [cite: 109]	0.146 MPa (146.84 kPa) [cite: 106, 107]	INFERIOR (-99.1%)
Wet Steel Adhesion Strength	28.3 MPa (4100 psi) [cite: 108, 109]	~0.150 MPa [cite: 106]	INFERIOR (-99.4%)
Compressive Strength	75.4 MPa (10,935 psi) [cite: 108]	0.05 - 0.20 MPa (Typical hydrogel) [cite: 113]	INFERIOR (-99.7%)
Handling Safety	Solvent-free, low-hazard paste [cite: 116, 117]	Class 6.1 Neurotoxin (Acrylamide) exposure risk [cite: 110, 121]	INFERIOR (Severe Risk)
Form Factor / Mixing	Simple 1:1 volume mix, paste [cite: 126]	Low-viscosity liquid requiring precise in-situ gelling [cite: 113]	INFERIOR (High wash-out risk)

METRIC	BELZONA 1212 (MARKET LEADER)	PAM-ALG-PDA HYDROGEL (VENTURE CONCEPT)	DELTA / SUPERIORITY
Retail Price	~\$377.00 / kg (€154 / 450g) [cite: 127]	~\$45.00 / kg (Est. market ceiling for non-structural sealants)	Cheaper, but functionally useless for structural applications

Conclusion on Superiority: The venture concept is entirely unsuited for the commercial marine adhesive market. A structural deficit of 117x on the primary metric cannot be overcome by pricing strategy or marketing.

Economic Profile

The economics block below calculates the cost structure. While the raw feedstocks (being predominantly water) are exceptionally cheap, the operational costs of handling toxic materials and the massive CapEx required for regulatory compliance destroy the financial model. Furthermore, because the product fails the primary functional test, time-to-first-revenue is essentially infinite for the target market.

CITED INPUT PRIMITIVES — EXACTLY WHAT THE CALCULATOR WAS GIVEN

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Honesty Clause Declaration (Instruction 4):

I explicitly state that this venture fails the baseline investment criteria.

1. **Delta on Primary Metric:** The performance delta is *negative* 99.1% compared to the incumbent [cite: 106, 109]. It is vastly below the 2x superiority threshold; it is functionally inferior by two orders of magnitude.

2. **CapEx:** The CapEx strictly exceeds \$250,000. Because the process requires compounding Acrylamide, a Class 6.1 toxic solid [cite: 110], standard commercial paint/epoxy blending equipment is illegal and unsafe to use without massive upgrades to dust extraction, sealed hoppers, and negative-pressure environmental controls [cite: 121].

3. **Time to Revenue:** Time to first paid delivery exceeds 18 months because no commercial marine engineer, port authority, or commercial diver will qualify or purchase a structural adhesive that fails at 0.146 MPa. The product would require a total pivot to biomedical wound-care markets (FDA Class II/III medical devices), which inherently require 3 to 7 years of clinical trials and regulatory approval.

Absence Audit

Why has the market ignored this "breakthrough"? The previous report claimed the market was ignoring a scientifically verified miracle due to a lack of translational vision. The reality is much simpler: **The market is ignoring this because it does not work for the intended application.**

In materials science literature, terms like "ultra-high-strength" are entirely relative to the material class being studied. When researchers describe a hydrogel with 146 kPa adhesion as "high-strength" [cite: 106, 107], they are comparing it to other hydrogels, biological tissues (like liver or skin), or weak hydrocolloid bandages. A marine engineer, however, defines "high-strength adhesive" in the context of steel, concrete, and fiberglass, where performance is universally measured in Megapascals (1 MPa = 1000 kPa).

The incumbent, Belzona 1212, is an epoxy novolac matrix packed with ceramic and aggregate fillers [cite: 116, 128]. It achieves extreme compressive strength (75.4 MPa) and pull-off adhesion (17.2 MPa underwater) because the epoxy matrix forms dense, irreversible, highly crosslinked covalent networks that are entirely impervious to water once cured [cite: 108, 109].

Conversely, a hydrogel is a swollen network of polymers containing upwards of 80% water. While the addition of Polydopamine (PDA) nanoparticles provides excellent interfacial catechol bonding (mimicking mussel foot proteins) [cite: 112, 118], the bulk material itself remains soft and compliant. You cannot hold two pieces of steel together under hydrodynamic stress using a material that has the mechanical consistency of a gummy bear, regardless of how tenaciously that gummy bear sticks to the steel surface. The cohesive failure of the bulk hydrogel will always precede the adhesive failure of the PDA-steel interface.

Therefore, the absence of this technology in the industrial marine sector is not a commercial gap; it is a fundamental physical limitation of the material class. The true commercial home for PDA-nanocomposite hydrogels remains strictly within the biomedical sector (e.g., wet tissue adhesives, wearable biosensors, and internal wound dressings) [cite: 106, 107], where compliance, softness, and biocompatibility are strictly required, and absolute mechanical shear strength is secondary.

Deterministic economics

Ultra-High-Strength Underwater Adhesives via In Situ Catechol-Polymer Co-deposition

Economics verdict: FAIL

DERIVED METRIC	VALUE
COGS per kg	\$5.98
Price per kg (gate basis = parity)	\$377.00
Venture's intended ask per kg	\$45.00
Incumbent price per kg	\$377.00
Price premium vs incumbent	-88.1%
Gross margin at parity	98.4%
Gross margin at the ask	86.7%
Contribution per kg	\$371.02
All-in OPEX per kg (itemised)	\$5.95
Gross margin, all-in OPEX basis	98.4%
Annual output (kg)	384,000
Annual revenue at nameplate (<i>capacity ceiling, assumes 100% sell-through</i>)	\$144,768,000
Annual gross profit at nameplate	\$142,471,837
Startup CapEx	\$350,000
Cash to first revenue (qualification)	\$150,000
Total cash at risk (CapEx + qualification)	\$500,000
Capital productivity (rev/CapEx)	413.62x
Breakeven volume (kg)	1,348
Payback from first sale (mo)	0.0
Payback incl. qualification wait (mo)	24.0
IRR (annualised, 60-mo horizon)	n/a — not meaningful (payback 0.0 mo — IRR unstable below 3 mo)

Formula definitions (LaTeX)

$$COGS\ per\ kg = \frac{\text{feedstock input cost} + \text{other variable cost}}{\text{conversion yield}} = 5.98\ USD/kg$$

$$GM_{parity} = \frac{P_{parity} - COGS}{P_{parity}} = 98.41\%$$

$$Contribution\ per\ kg = P_{parity} - COGS = 371.02\ USD/kg$$

$$Capital\ productivity = \frac{annual\ revenue}{startup\ CapEx} = 413.62\times$$

$$Breakeven\ volume = \frac{startup\ CapEx}{contribution\ per\ kg} = 1347.63\ kg$$

$$Payback\ from\ first\ sale = \frac{total\ cash\ at\ risk}{monthly\ gross\ profit} = 0.04\ months$$

Minimum viable equipment (sourced, itemised)

ITEM	SPEC	NEW (\$)	USED (\$)	VENDOR / WHERE
Class 6.1 Rated Reactor	1000L Stainless Steel with enclosed powder hopper and fume extraction	\$250,000	\$120,000	Industrial Mixing Solutions USA
Nitrogen Purge System	High purity continuous flow gas delivery	\$50,000	\$25,000	AirGas USA
Hazmat Packaging Line	Automated fill for Class 6.1 liquid precursors	\$50,000	\$20,000	PackMach

CapEx total **\$\$\$350,000** vs sum of line items **\$\$\$350,000**: RECONCILES.

All-in OPEX per unit (itemised)

COMPONENT	COST PER UNIT
feedstock	\$1.45
energy	\$0.50
labor	\$2.00
water	\$0.05
maintenance	\$0.45
waste_disposal	\$0.50
packaging	\$1.00

Sum **\$\$\$5.95/unit**. Components reconcile to the stated total.

⚠ Capital productivity of 414x is not a return — it is a signal that capital is no longer the binding constraint. At this level the limiting factor is whether 384,000 kg/yr can actually be SOLD.

Treat annual revenue as a capacity ceiling and verify it against the report's own SAM before believing any of it. The low CapEx is real; the revenue is a hypothesis.

Threshold checks

CHECK	VALUE	RESULT
Gross margin $\geq 70\%$	98.4%	PASS
Startup CapEx $\leq \$250,000$	\$350,000	FAIL
Payback ≤ 12 mo	0.0 mo	PASS
Capital productivity $\geq 2.0x$	413.62x	PASS
Price parity ($\leq 10\%$ premium)	-88.1%	FAIL

Sensitivity (does it survive being wrong?)

SCENARIO	GROSS MARGIN	PAYBACK (MO)	IRR	CAP. PRODUCTIVITY
base	98.4%	0.0	n/m	413.62x
price -25%	98.4%	0.0	n/m	413.62x
yield -25%	98.3%	0.0	n/m	413.62x
CapEx +100%	98.4%	0.1	n/m	206.81x
feedstock +50%	98.2%	0.0	n/m	413.62x
stacked (price -25%, yield -25%, CapEx +100%)	98.3%	0.1	n/m	206.81x

Assumptions: gross profit only (no SG&A/working capital), nameplate utilisation from month of first revenue, qualification spend amortised evenly over the wait, 60-month horizon, no terminal value. IRR is a ranging device, not a forecast.

The case against it

Ultra-High-Strength Underwater Adhesives via In Situ Catechol-Polymer Co-deposition

- Rubric verdict: FAIL
- **Audit verdict: FAIL**
-  **WARN / duplicate_refs** — Cites duplicate reference(s) [10, 14, 49] that restate a source already counted, inflating the apparent evidence base.
-  **WARN / declared_parity** — Price parity is DECLARED, not demonstrated: product and incumbent price are both 377.0 citing the same ref (16). The parity check cannot fail when one number is written twice; verify the incumbent price against an independent market source.

Institutional capital-formation pack

Ultra-High-Strength Underwater Adhesives via In Situ Catechol-Polymer Co-deposition

Purchase profile: oneoff — working-capital cycle: 45 days. The ramp curve for a oneoff purchase pattern is 40%, 65%, 85%, 95%, 100% of nameplate (\$144,768,000 annual revenue at capacity).

Pro-forma P&L, cash flow and debt service (5 years)

YEAR	REVENUE	GROSS PROFIT	EBITDA	INTEREST	PRINCIPAL	TAX	FCF	DSCR
Y1	\$57,907,200	\$56,988,735	\$48,302,655	\$60,000	\$73,467	\$10,143,557	\$38,025,631	285.91x
Y2	\$94,099,200	\$92,606,694	\$78,491,814	\$51,184	\$82,283	\$16,483,281	\$61,875,066	464.60x
Y3	\$123,052,800	\$121,101,061	\$102,643,141	\$41,310	\$92,157	\$21,555,060	\$80,954,615	607.55x
Y4	\$137,529,600	\$135,348,245	\$114,718,805	\$30,251	\$103,215	\$24,090,949	\$90,494,389	679.03x
Y5	\$144,768,000	\$142,471,837	\$120,756,637	\$17,865	\$115,601	\$25,358,894	\$95,264,276	714.77x

Minimum DSCR over the term: 285.91x — free cash flow turns positive in year 1.

Debt structure and amortization

Senior amortizing term loan: principal **\$500,000**, 5-year term, 12.0% APR, monthly annuity payment **\$11,122** (total debt service \$133,467/yr). Sized so the worst year still clears a 1.25x DSCR, and capped at 80% of hard assets + working capital (\$18,198,110).

YEAR	OPENING BALANCE	PAYMENT	INTEREST	PRINCIPAL	CLOSING BALANCE
Y1	\$500,000	\$133,467	\$60,000	\$73,467	\$426,533
Y2	\$426,533	\$133,467	\$51,184	\$82,283	\$344,251
Y3	\$344,251	\$133,467	\$41,310	\$92,157	\$252,094
Y4	\$252,094	\$133,467	\$30,251	\$103,215	\$148,879
Y5	\$148,879	\$133,467	\$17,865	\$115,601	\$33,277

Use of proceeds: startup CapEx \$350,000 · qualification/pre-revenue spend \$150,000 · working capital \$17,848,110 · total raise **\$500,000**.

Bankability

Verdict: BANKABLE (model) — min DSCR 285.91x.

- contracted-floor sensitivity raises min DSCR from 285.91 to 543.22 — an offtake/PPA-style floor materially improves bankability

Contracted-revenue sensitivity (the PPA test)

With 60% of nameplate revenue under a multi-year offtake contract from year 1, debt capacity stays at the use-of-proceeds cap (\$500,000), but min DSCR headroom rises from 285.91x to **543.22x**.

Contracted revenue is what converts a good margin into a bankable cash flow: it is the single most valuable document in the capital-formation file.

Valuation (derived from the pro-forma)

METHOD	VALUE
DCF @ 15%, no terminal value	\$232,184,821
DCF @ 25%, no terminal value	\$179,752,010
DCF @ 15%, terminal 4x EBITDA	\$472,334,383
DCF @ 25%, terminal 4x EBITDA	\$338,030,148
Revenue multiple 2x–4x on Y5	\$289,536,000 – \$579,072,000

Indicative enterprise value band: \$179,752,010 – \$579,072,000. This is a ranging device computed from the pro-forma, not a negotiated price — a tokenization platform will re-value independently.

Token structure recommendation

Amortizing debt token (bond-like). min DSCR 285.91 >= floor 1.25 and the debt capacity (500,000 USD, 100% of the raise) is material — fixed coupon, fixed maturity, monthly principal amortization is the structure the cash flow supports.

Morpho Blue LLTV recommendation: 77.0% (from the governance-approved set 0%, 38.5%, 62.5%, 77%, 86%, 91.5%, 94.5%, 96.5%, 98%) — bankable cash flow at the stated assumptions supports the standard commercial band (77% of the approved set); higher LLTVs trade liquidation margin for leverage.

Maximum sustainable annual borrow cost: 7631.8% — the worst-year after-tax EBITDA expressed against the debt principal. Borrowing above this rate inverts coverage at the stated assumptions.

Platform due-diligence readiness

Brickken — items below marked *covered* are answered by this dossier (report section or this pack); items marked *operator must supply* are legal/regulatory documents only the venture operator can produce:

DD ITEM	STATUS
Corporate entity & KYB/KYC pack	operator must supply
Business plan & market evidence	covered by the report
Financial statements / projections (3-5y)	covered by this pack

DD ITEM	STATUS
Independent valuation & pricing rationale	covered by this pack
Tokenomics design (supply, class, rights)	covered by this pack
Legal review of the token instrument	operator must supply
Smart-contract security audit	independent third party
Marketing & investor-acquisition plan	covered by the report

Centrifuge — items below marked *covered* are answered by this dossier (report section or this pack); items marked *operator must supply* are legal/regulatory documents only the venture operator can produce:

DD ITEM	STATUS
Pool type chosen (Closed/Portfolio avoids the POP governance vote)	covered by this pack
SPV / asset-originator entity (bankruptcy-remote, true sale)	operator must supply
POP-grade issuer pack (team, named legal/accounting partners, 2y origination history)	operator must supply
Asset valuation & NAV methodology (DCF mark-to-model)	covered by this pack
Cash-flow projections + covenant reporting	covered by this pack
Senior/junior tranche structure + subordination ratio	covered by this pack
Independent auditor / trustee / servicer	independent third party
Investor KYC/AML (ID, W-8BEN/W-9, accredited-investor check)	operator must supply

Centrifuge expects institutional scale for OPEN pools ($\geq$ \$25M at launch, $\geq$ \$100M within 12 months ideal, ~\$5M first-loss junior, senior pre-committed). A micro-venture should structure as a CLOSED or PORTFOLIO pool, which skips the governance vote entirely. One SPV per pool, assets true-sold to it.

Morpho Blue — items below marked *covered* are answered by this dossier (report section or this pack); items marked *operator must supply* are legal/regulatory documents only the venture operator can produce:

DD ITEM	STATUS
Collateral/loan asset pairing (ERC-20s; loan asset not ERC-4626)	operator must supply
Oracle availability & quality for the pair (immutable at creation)	operator must supply
LLTV from the governance-approved set, with rationale	covered by this pack
Liquidation depth — liquidators must be able to sell the collateral (LIF economics)	operator must supply
Curator path: caps, timelocks $\geq$ 3 days, interface listing policy	covered by this pack
RWA route: tokenized equity/debt is not directly usable — enter via Centrifuge vault tokens	covered by this pack
Underlying cash-flow coverage of borrow cost	covered by this pack

Morpho Blue markets are permissionless and immutable: one collateral asset, one loan asset, an LLTV from the approved set (0 / 38.5 / 62.5 / 77 / 86 / 91.5 / 94.5 / 96.5 / 98%), a fixed oracle, and a governance-approved IRM. An illiquid RWA token cannot be posted as collateral directly — the realistic financing path is: Brickken/Centrifuge issuance -> vault token -> Morpho Blue market. This pack's LLTV recommendation assumes that route.

Checklist basis — Brickken: tokenization onboarding (KYB/KYC, corporate docs, financials, business plan, valuation, legal review, smart-contract audit) per brickken.com; EU framework: Regulation (EU) 2023/1114 (MiCA), Prospectus Regulation (EU) 2017/1129, Securitisation Regulation (EU) 2017/2402; smart-contract audit practice per ConsenSys Diligence (consensys.net/diligence). Centrifuge: pool types + POP v3 (CP68), legal offering / SPV true-sale, multi-tranche + subordination breach, pool valuation & NAV, investor onboarding per docs.centrifuge.io and gov.centrifuge.io. Morpho Blue: market parameters, oracle, liquidation incentive factor, curator & vaults per docs.morpho.org. Project-finance bankability: DSCR floor and debt sizing conventions from solar PPA / project-finance practice.

Model assumptions (one table, every concept)

ASSUMPTION	VALUE	PROVENANCE
Nameplate revenue / gross margin	\$144,768,000 / 98.4%	MEASURED (report economics block)
Revenue ramp by purchase frequency (oneoff)	40%, 65%, 85%, 95%, 100% of nameplate	ASSUMED (pack default)
SG&A	15% of revenue	ASSUMED (pack default)
Tax	21% flat, losses carried forward	ASSUMED (pack default)
Debt	5y, 12.0% APR, monthly annuity	ASSUMED (pack default)
DSCR floor / LTV cap	1.25x / 80%	ASSUMED (institutional convention)

Everything not marked MEASURED is a declared model assumption, identical across all concepts so comparisons are like-for-like. No figure in this pack is invented per-concept; every number is either the report's cited primitive or arithmetic on it.

Pro-forma, debt and valuation figures are computed deterministically from this report's cited economics primitives (institutional.py). Provenance: MEASURED = the report's cited primitive; DERIVED = arithmetic on measured values; ASSUMED = a declared pack default, identical across every concept.

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86 unique sources for this concept. Every quantitative claim traces to one of these; check them.

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